



Dhyey Patel

Bachelor of Technology
in Electronics and Communication Engineering
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PROFESSIONAL SUMMARY

Passionate about building intelligent systems that solve real-world problems using Machine Learning, Deep Learning, Computer Vision, and Large Language Models (LLMs). Experienced in developing AI applications, including deepfake detection systems and agentic AI solutions, with hands-on knowledge of Python, PyTorch, TensorFlow, and the end-to-end machine learning workflow. Always eager to learn new technologies, tackle challenging problems, and contribute to impactful AI solutions while growing as an AI professional.

EDUCATION

B.Tech in Electronics and Communication Engineering (ECE) **2024–2028**
Sardar Vallabhbhai National Institute of Technology, Gujarat
CGPA: 8.04 (End of 4th Semester)

CBSE (Class XII) **2024**
Shree Swaminarayan International School, Gujarat
Percentage: 96.4%

CBSE (Class X) **2022**
Shree Shreeji English Medium School, Gujarat
Percentage: 94%

PROJECTS

- **DeepFake Detection System | Python, PyTorch, CNN, Gradio** *Live Demo*
GitHub
 - Developed a CNN-based deep learning model to detect deepfake images.
 - Performed image preprocessing, feature extraction, and dataset augmentation to improve model performance.
 - Trained and evaluated the model on real and manipulated datasets for accurate binary classification.
 - Analyzed facial inconsistencies and visual artifacts to distinguish authentic and manipulated content.
- **Advanced RAG Document Analyst | SentenceTransformers, CrossEncoder, BM25, ChromaDB** GitHub
 - Built a modular Retrieval-Augmented Generation (RAG) system to process complex PDFs using hybrid search (Dense Embeddings + Sparse BM25) fused with Reciprocal Rank Fusion (RRF).
 - Enhanced retrieval accuracy by implementing advanced query preprocessing techniques, including Multi-Query Expansion and Hypothetical Document Embeddings (HyDE).
 - Integrated Cross-Encoder reranking (ms-marco-MiniLM-L-6-v2) and Google Gemini API for context-aware, streaming responses with exact inline source citations.
 - Containerized and deployed the application as an interactive Streamlit UI on Modal.com with real-time pipeline execution tracing.
- **Sidekick — Autonomous AI Co-worker | LangChain, LangGraph, Google Gemini, Playwright, MCP, Python** GitHub
 - Built an autonomous AI agent for multi-step web research, workflow automation, and document generation using LangChain, LangGraph, and Google Gemini.
 - Implemented an Evaluator–Worker reflection loop for iterative validation and automated refinement based on user-defined success criteria.
 - Integrated Playwright and Model Context Protocol (MCP) for persistent browser automation, web interaction, and sandboxed filesystem access.
 - Developed security guardrails including PII masking, Human-in-the-Loop approvals, error recovery, live execution planning, and push notifications.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++
- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, Generative AI, Large Language Models (LLMs), AI Agents, Prompt Engineering, Retrieval-Augmented Generation (RAG)
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, Spring, SpringBoot
- **Tools & Platforms:** Git, GitHub, Visual Studio Code, Cursor, Jupyter Notebook, Gradio, Hugging Face, Kaggle, Playwright, Model Context Protocol (MCP)
- **Concepts:** Model Fine-tuning, Transfer Learning, NLP, Vector Databases, Web Automation, REST APIs